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Set 8 Employee Performance Analysis

Dataset: Employee Performance

Employee ID Department Years of Service Performance Score

1 Sales 5 85

Employee ID Department Years of Service Performance Score

2 HR 3 92

3 Marketing 7 78

1. Create a line chart to visualize the performance trend of employees over time.

Label the axes and

title the chart.

2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

3. Build a table to display the performance data for each employee. Label the table elements.

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

SOLUTION :

CODE :

01.

%%R

```
library(ggplot2)
```

```
df8 <- data.frame(  
  Employee_ID = factor(c(1, 2, 3)),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)
```

```
df8_sorted <- df8[order(df8$Years_of_Service), ]
```

```
ggplot(df8_sorted, aes(x = Years_of_Service, y = Performance_Score, group =
```

```

1)) +
  geom_line(color = "#2B5C8F", size = 1.2) +
  geom_point(color = "#D9534F", size = 4) +
  geom_text(
    aes(label = paste0(Department, "\n(", Performance_Score, ")")),
    vjust = -0.8,
    size = 3.8,
    fontface = "bold"
  ) +
  labs(
    title = "Employee Performance Trend over Years of Service",
    x = "Years of Service",
    y = "Performance Score",
    caption = "Data Source: Set 8 Employee Performance"
  ) +
  scale_y_continuous(limits = c(70, 100)) +
  theme_minimal()

```

02 %%R

```
library(ggplot2)
```

```

df8 <- data.frame(
  Employee_ID = factor(c(1, 2, 3)),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

```

```

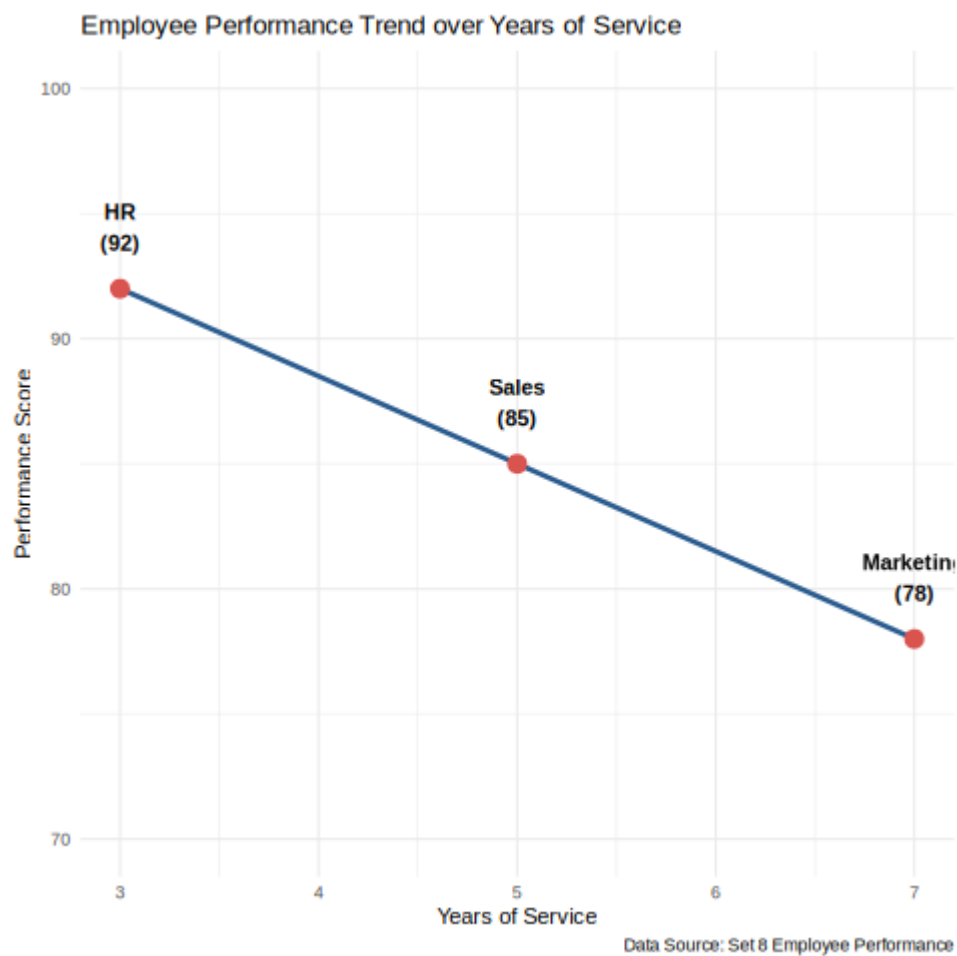
ggplot(df8, aes(x = Department, y = Performance_Score, fill = Department)) +
  geom_bar(stat = "identity", width = 0.5) +
  geom_text(
    aes(label = paste0("Score: ", Performance_Score)),
    vjust = -0.5,
    size = 4,
    fontface = "bold"
  ) +
  labs(
    title = "Employee Performance Score by Department",
    x = "Department",
    y = "Performance Score",
    fill = "Department"
  ) +
  theme_minimal() +
  scale_fill_brewer(palette = "Set2")

```

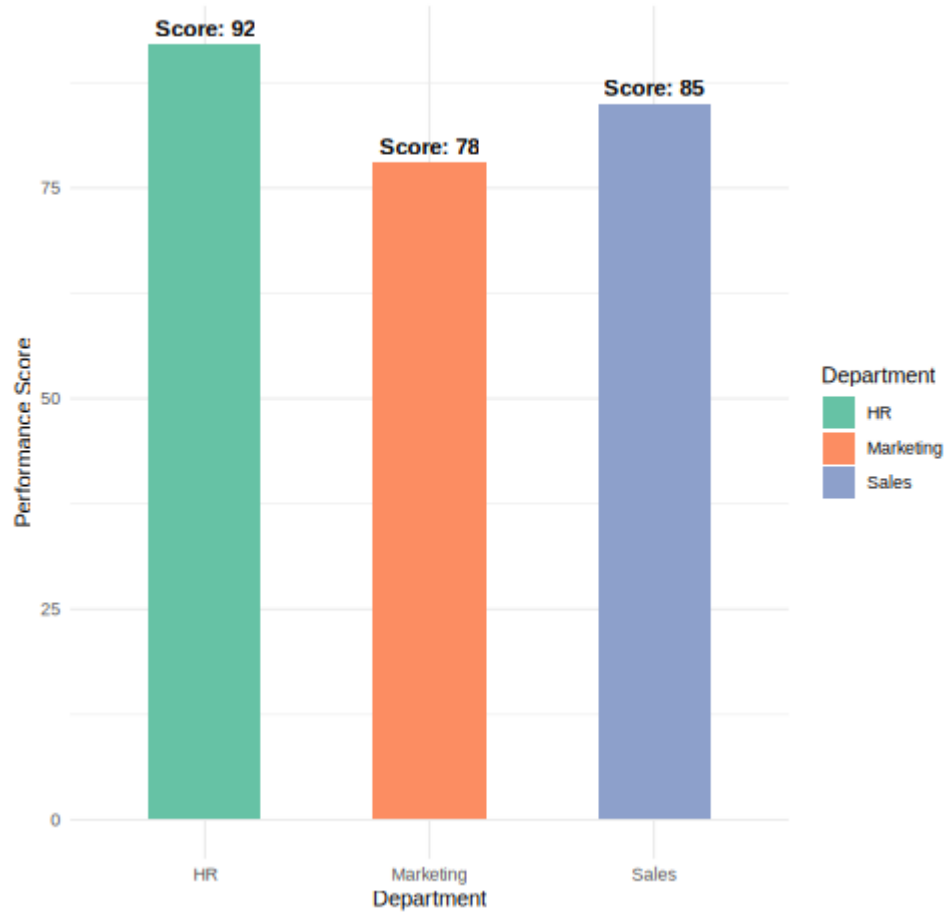
03. %R

```
df7 <- data.frame(  
  Customer_ID = c(1, 2, 3),  
  Age = c(28, 35, 42),  
  Gender = c("Female", "Male", "Female"),  
  Income_USD = c(50000, 60000, 75000)  
)  
  
df7_table <- df7  
df7_table$Income_USD <- paste0("$", format(df7_table$Income_USD,  
big.mark = ","))  
  
print("--- CUSTOMER DEMOGRAPHICS TABLE ---")  
print(df7_table)
```

OUTPUT :



Employee Performance Score by Department



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